

Pranav Kumar Soma

SOFTWARE ENGINEER

prsama@ucsd.edu • github.com/soma-p • LinkedIn • pranavsoma.me • San Diego / SF Bay Area

EDUCATION

M.S. Computer Science, UC San Diego

Dec 2026 (expected)

GPA 4.00 • Jacobs School of Engineering • M.S. in 1.25 years

B.S. Computer Science, UC San Diego

Dec 2025

GPA 3.96 • Cum Laude, CSE Department Honors, Provost Honors (all quarters) • B.S. in 2 years

Coursework: Algorithms, Operating Systems, Theory of Computation, Software Engineering, Data Systems for ML, Search & Optimization, Computer Vision, Statistical NLP, Discrete/Convex Optimization.

EXPERIENCE

Founding Software Engineer | Stealth Startup

Starting Sep 2026

Autonomous systems • part-time

- Building the **core operating system & software infrastructure** from the ground up: hardware-agnostic, fault-tolerant, horizontally scalable
- Owning **distributed messaging & networking** for edge nodes on degraded links, low-latency **C2** operator dashboards, and the **simulation + hardware-in-the-loop** harness the team tests real systems against

Software Engineer Intern | ServiceNow

Summers 2025 & 2026

AI Voice Agents, IT Service Management • **Technical Skills Award** • top intern of 800+

- Cut **AI Voice Agent latency 45%** by re-architecting the NLU pipeline onto higher-capacity models across 8 intents; shipped to production and **validated live with an enterprise customer. Return offer; invited back for a second summer**
- Delivered the AWS Lex V1→V2 migration ahead of schedule, unblocking every ITSM voice deployment before end-of-life; shipped a post-incident “wrap-up” modal (UI + Express backend routing)

AI Research Fellow & Agentic Systems Lead | Laboratory for Emerging Intelligence, UCSD

Apr 2025–Present

- Lead **Teradata-funded** agentic-AI platform serving 7,000+ users · custom **MCP servers** over UCSD APIs + a deterministic, fault-graceful workflow executor
- Built & shipped Python **AI Tutor** (deployed at UCSD, SDSU, Cal Poly Pomona) for **8,000 students / 8 courses**; FastAPI services, MongoDB · **64% fewer TA hours**

Honors & M.S. Thesis Researcher | UCSD CSE · Computational Redistricting

Sep 2024–Present

- Engineered end-to-end geospatial pipeline scoring U.S. redistricting plans over Census-2020 data (**158K–5.5M blocks/state**, 5–18M-row Parquet store)
- Built 3 multi-threaded generators (greedy DP + Kernighan–Lin, Sequential Monte Carlo, XGBoost + multilevel graph partitioning); Python, pymetis, GeoPandas

SELECTED PROJECTS

- ORCA · AI Music Production** (*Best Project, SDTC: 1 of 30 from 200+; SAIRS 2025*) · full-stack app; OpenCV + neural source separation, CNN audio-to-score; **98% accuracy**. Flask, Docker, React
- Spay.LA** (*spay.la*) · MERN fundraising & legislative platform; **raised \$75K**; led 11 developers as Engineering Manager. Firebase/Google auth
- AgentSpace** · hireable expert agents over two-way **GraphRAG** on skill embeddings, served over **MCP + A2A**

LEADERSHIP

Co-Founder & President, Real World AI Network (2026–Present) · founded **7,000+ student** federation across UCSD’s engineering, data-science & business schools driving AI research & entrepreneurship.

Founding Board Member & VP External Affairs, University AI Alliance · student-led frontier-tech network spanning MIT, Stanford, Caltech & Cornell; own external relationships with a16z speedrun, Bain Capital Ventures, Techstars & execs at Google and Meta.

President, CSE Society @ UCSD (200+ members, 21+ projects) • **Founder**, Innovate Research Group (30+ across UCLA, UCSB, UCSC) •

Awards: YC Startup School 2026, National Merit Finalist, USACO Silver.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, Swift, Kotlin, SQL, ARM-32 Assembly, HTML/CSS, Bash

Web & Full-Stack: React, Node.js, Express, FastAPI, Flask, Django & Django REST, MERN, Firebase, Google OAuth, REST APIs

Systems, Cloud & Data: AWS (Lex, Lambda), Docker, Git, Linux; MongoDB, PostgreSQL, MySQL; NumPy, Pandas, GeoPandas; parallel & multi-threaded computing, distributed & real-time systems, system design, finite state machines

ML & Enterprise: PyTorch, scikit-learn, XGBoost, OpenCV; LLM fine-tuning (LoRA, SFT, RLHF), RAG & GraphRAG, agents & MCP, LLM-as-judge evals, NLP/NLU, computer vision; ServiceNow (ITSM, CSM, SPM), Smartsheet, Agile/Scrum, ROS, embedded systems